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11 April 1985
SUPERSEDING
MIL-D-3898A
10 September 1959
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9 April 1963
(See 6.6)

MILITARY SPECIFICATION

DRAFTING MACHINE AND SCALES
GENERAL SPECIFICATION FOR

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers drafting machines and includes scales used to transfer a straight line at any angle, from any part of a drawing or chart, so that it will remain parallel to its original direction.

1.2 Classification. The drafting machines shall be of the following types, classes and styles as specified (see 6.2.1).

Type I - Plotting

Class 1 - Large plotting
Class 2 - Small plotting

Type II - Drafting

Class 1 - General drafting
Class 2 - Civil engineer
Class 3 - Detailers

Style A - Band drive
Style B - Rigid bar
Style C - Parallelogram guide rod

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Naval Sea Systems Command, SEA 55Z3, Department of the Navy, Washington, DC 20362-5101 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 6675

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2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. Unless otherwise specified, the following specifications and standards of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

PPP-B-566 - Box, Folding, Paperboard.
PPP-B-636 - Boxes, Shipping, Fiberboard.
PPP-B-676 - Boxes, Set-up.
PPP-C-1752 - Cushioning Material Packaging, Unicellular Polyethylene Foam, Flexible.

MILITARY

MIL-M-14 - Molding Plastics and Molded Plastic Parts, Thermosetting.
MIL-P-116 - Preservation, Methods of.
MIL-S-901 - Shock Tests, H.I. (High-Impact), Shipboard Machinery, Equipment and Systems, Requirements for.
MIL-A-8625 - Anodic Coatings, for Aluminum and Aluminum Alloys.

(See Supplement 1 for list of applicable specification sheets).

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.

2...2 Government drawings. The following Government drawings form a part of this specification to the extent specified herein.

DRAWINGS

NAVAL SEA LOGISTICS SUPPORT ENGINEERING ACTIVITY, MECHANICSBURG

9240-032 - Drafting Machine, Large Plotting, MK. 3, MOD. 4.
9240-033 - Drafting Machine, Small Plotting, MK. 4, MOD. 2.

(Copies of specifications, standards and drawings required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Other publications. The following document forms a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

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UNIFORM CLASSIFICATION COMMITTEE, AGENT
Uniform Freight Classification Ratings, Rules and Regulations

(Application for copies should be addressed to the Uniform Classification Committee Agent, Tariff Publication Officer, Room 1106, 222 South Riverside Plaza, Chicago, IL 60606.)

(Industry association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 Specification sheets. The individual item requirements shall be as specified herein and in accordance with the applicable specification sheets. In the event of any conflict between requirements of this specification and the specification sheet, the latter shall govern.

3.2 First article. When specified (see 6.2.1), sample units shall be subjected to first article inspection (see 4.3 and 6.3).

3.3 Material.

3.3.1 Drafting machine (less scales).

3.3.1.1 Metal parts. Unless otherwise specified, all metal parts used in the drafting machine shall be of corrosion resisting materials or other materials treated in a manner to render them adequately resistant to corrosion when tested in accordance with 4.6.2. Contact between dissimilar metals which may cause electrolytic action shall be avoided.

3.3.1.2 Plastic parts. Unless otherwise specified, plastic parts of the drafting machine shall conform to type CFI-20 of MIL-M-14.

3.3.2 Scales. Scales shall conform to the material requirements as specified in the applicable specification sheet (see 3.1).

3.4 Design.

3.4.1 Type I - plotting.

3.4.1.1 Class 1 - large plotting. The type I, class 1 drafting machine shall be in accordance with the requirements specified on Drawing 9240-032.

3.4.1.2 Class 2 - small plotting. The type I, class 2 drafting machine shall be in accordance with the requirements specified on Drawing 9240-033.

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3.4.2 Type II - drafting. The type II drafting machine shall be in accordance with the best commercial practices for drafting machine design to insure versatility and accuracy in operation with a minimum amount of effort. The drafting machine shall be furnished in style A, B or C as specified (see 6.2.1).

3.4.2.1 Protractor. The drafting machine protractor shall have a means provided for setting the azimuth (baseline or zero position of the protractor head) independently of the parallel motion device throughout 360 degrees. The protractor markings shall be graduated as specified herein and shall be oriented with 0 degree marking at the right horizontal and 90 degrees marking at the top vertical when viewed from the operator's position. Each 10 degree interval shall be designated with a numeral. The numerals marking the quadrants shall be in line with the respective radial angle and be readable from the operator's position.

3.4.2.1.1 Type II, class 1 general drafting machine protractor. The type II, class 1 general drafting machine protractor shall be graduated in single degree markings from 0 to 90 degrees. Numerals shall be in a single row on the protractor by quadrants. A double vernier graduated to read to 5 minutes of arc shall be provided with numerals designating each 15 minutes on the vernier.

3.4.2.1.2 Type II, class 2 civil engineer drafting machine protractor. The type II, class 2 civil engineer drafting machine protractor shall be graduated in 1/2 degree markings. Numerals shall be in two rows of figures, one by quadrants and one from 0 to 360 degrees. The 0 to 360 degree numerals shall be marked in 10 degree intervals with the top of the numeral toward the centerpoint. A double vernier graduated to read 1 minute of arc shall be provided with numerals designating each 15 minutes on the vernier.

3.4.2.1.3 Type II, class 3 detailers drafting machine protractor. The type II, class 3 detailers drafting machine protractor shall be graduated in single degree markings from 0 to 90 degrees. Numerals shall be in a single row on the protractor by quadrants. A double vernier graduated to read 5 minutes of arc shall be provided with numerals designating each 10 minutes on the vernier.

3.4.2.2 Parallel motion device.

3.4.2.2.1 Style A band drive drafting machine. The style A band drive drafting machine obtains parallel motion through the use of continuous one-piece steel bands passing over the pulleys. The bands shall be easily replaceable in the event of breaking. The bands shall have a protective cover to avoid injury to the operator. Means shall be provided for easy adjustment of the band tension.

3.4.2.2.2 Style B rigid bar drafting machine. Style B rigid bar drafting machine obtains parallel motion through the use of a rigid bar or vertical rail suspended from a horizontal carriage for supporting the protractor head. The bars shall be easily replaceable in the event of damage. Means shall be provided for independently locking the horizontal and vertical motion of the machine. Means shall be provided for adjustment of horizontal and vertical accuracy.

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3.4.2.2.3 Style C parallelogram guide rod drafting machine. Style C parallelogram guide rod drafting machine obtains parallel motion through the use of guide rods so arranged as to function in the manner of a parallelogram. The guide rods shall be easily replaceable in the event of damage.

3.4.2.3 Scale arm. The design of the scale arm for the type II drafting machine shall be such as to hold two scales at 90 degrees to each other. The scale arm shall be resilient, yet hold the scales firmly and allow the scales to be quickly and easily removed. The chuck end of the scale arms shall be capable of receiving and retaining chuck plate as specified in MIL-D-3898/1.

3.4.2.4 Adjustments.

3.4.2.4.1 Protractor head. The protractor head shall be capable of being adjusted or locked by means of a positive clamping mechanism, in any desired position in azimuth either with or independently of the scale arm.

3.4.2.4.2 Micrometer adjusting screw. When specified (see 6.2.1), a type II, class 2 drafting machine shall be supplied with a micrometer adjusting screw to provide for fine adjustment of angles after the instrument has been oriented.

3.4.2.5 Automatic indexing. The type II, class 1 and 3 drafting machines shall provide for positive locking of the protractor arm at either a 3- or 15-degree position as specified (see 6.2.1), throughout the entire circle. Indexing shall be by means of a detent action or similar indexing or escapement action allowing for free rotation and setting of intermediate angles. If index detent is positioned at the right horizontal of protractor head, the 0 degree marking on protractor (see 3.4.2.1) may be located up to 20 degrees above or below right horizontal position.

3.4.2.6 Anchor. The anchor for the drafting machine shall be of simple design to permit the drafting machine to be easily mounted in a positive manner by being screwed, bolted, or clamped in place or suspended from a track mounted on the full length of the top of the board. The anchor shall permit the drafting head to be raised to a height of at least 4 inches from the board. Means shall be provided for securely attaching the anchor or carriage to either a wooden or metal table or board as specified (see 6.2.1).

3.4.2.7 Arm size. The drafting machine shall be furnished in one of the following sizes as specified (see 6.2.1).

Type II, classes 1 and 2 - 24-, 30-, or 36-inch arms.
Type II, class 3 - 16- or 18-inch arms.

3.4.2.8 Counterbalancing. When specified (see 6.2.1), means shall be provided for counterbalancing the drafting machine when used on a steep inclined board or table.

3.4.3 Scales. The drafting machine scales shall be in accordance with the applicable specification sheet (see 3.1).

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3.4.3.1 Graduations and markings. Scales shall have graduations and markings conforming to the arrangement, size, and tolerances as specified herein and on the applicable specification sheet (see 3.1) when examined in accordance with 4.5.2.1. Graduations and markings shall be stamped, engraved, or molded, and shall be 0.015-inch deep by 0.010-inch wide. Unless otherwise specified (see 6.2.1), letters and numerals shall be 5/64-inch high. Graduations and markings shall be black except as specified herein. The marking material shall be insoluble in water and shall bond or fuse permanently to the scale. Graduations and markings on plastic scales shall be on the underneath side in order that they may be correctly read from the top side of the scale. The graduations on plastic scales shall be protected from abrasion and wear by either a 0.005-inch transparent plastic lamination or raising the under surface of the edge 0.005-inch minimum so that they will be clear of the drawing surface.

3.4.3.1.1 Range-time-speed plotting scale. The graduations and markings on the scales (see MIL-D-3898/6) shall be black, except that the inner row (reciprocal scale) of numerals on the yards scale shall be magenta in color.

3.4.3.1.2 Tolerances. Tolerances for spacing of all graduations shall apply at a temperature of 23 degrees Celsius (°C) and relative humidity of 50 percent.

3.4.3.1.2.1 Aluminum. Spacing between all graduations less than 1-inch apart shall be within plus or minus 0.002 inch of that specified. Spacing between graduations more than 1-inch apart shall be within plus or minus 0.008 inch per foot of graduation.

3.4.3.1.2.2 Plastic and boxwood. Spacing between all graduations less than 1-inch apart shall be within plus or minus 0.005 inch of that specified. Spacing between graduations more than 1-inch apart shall be within plus or minus 0.015 inch per foot of graduation.

3.4.3.1.3 Identification markings. The scale shall be marked with the applicable part number on the underside of the scale at the location specified on the applicable specification sheet. Markings shall be stamped, engraved, or molded in order that they may be correctly read from the underside of the scale. Letters and numerals shall be 1/16-inch high and filled with permanent black paint.

3.4.3.2 Material protection.

3.4.3.2.1 Aluminum. Aluminum scales shall be protected with an anodic coating in accordance with MIL-A-8625.

3.4.3.2.2 Boxwood. Boxwood scales shall be faced with suitable white plastic.

3.5 Performance.

3.5.1 Parallelism. The drafting machine shall maintain parallelism within 0.08-inch over an 18-inch length when tested in accordance with 4.5.1.2.1.

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3.5.2 Automatic indexing. The automatic indexing of type II, class 1 and 3 drafting machines shall be accurate within plus or minus 10 minutes at each of the detent positions (see 3.4.2.5) when tested in accordance with 4.5.1.2.2.

3.5.3 Protractor accuracy. The accuracy of the protractor shall be within plus or minus 10 minutes when tested in accordance with 4.5.1.2.3.

3.5.4 High impact shock. The type I drafting machine shall display no hazard to personnel, and meet the requirements specified herein, after testing in accordance with 4.6.1.

3.6 Physical characteristics.

3.6.1 Scale flatness. The deviation from flatness shall not exceed 0.060 inch at the center of the scale when tested in accordance with 4.5.2.2.1.

3.6.2 Scale edges. Ruling edges of the scale shall be smooth, straight, and parallel within 0.01 inch when tested in accordance with 4.5.2.2.2 and 4.5.2.2.3.

3.7 Storage. The drafting machine, less scales, shall show no evidence of damage affecting serviceability or appearance when subjected to a temperature of plus 71°C and minus 62°C in accordance with storage tests specified in 4.6.2.

3.8 Accessory equipment.

3.8.1 Tools. Each drafting machine will be furnished with any special tools required for scale removal or any additional special tools needed to make adjustments to the drafting machine.

3.8.2 Carrying or storage case. Unless otherwise specified (see 6.2.1), each drafting machine shall be furnished with a carrying or storage case. The case shall be a hardwood or metal case fitted with necessary securing chocks to receive and hold securely the instrument and accessories.

3.9 Marking. The drafting machine and carrying case shall have the following marking permanently attached. The marking shall be either directly engraved or shall be suitably reproduced on a separate plate:

(a) Drafting machine MIL-D-3898.
(b) Type ___ class ___ style___.
(c) Name of contracting activity.
(d) Manufacturer's name.
(e) Contract no.
(f) Year.
(g) U.S.

3.10 Manuals. Each drafting machine shall be furnished with operating instructions in accordance with the contractor's standard instruction literature for similar equipment.

3.11 Workmanship. Workmanship shall be in accordance with good commercial practices for this type of product and shall be free from nicks, scratches and burrs.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of inspection. The inspections specified herein are classified as follows:

(a) First article inspection (see 4.3).
(b) Quality conformance inspection (see 4.4).

4.3 First article inspection. First article inspection shall consist of the examination and tests specified in 4.5 and 4.6.

4.3.1 First article inspection report. A first article test report shall be provided in accordance with the data ordering document (see 6.2.2).

4.4 Quality conformance inspection. Quality conformance inspection shall consist of a sample number of complete machines (with scales) or of scales only selected in accordance with 4.4.1 and shall be subjected to the examinations and tests specified in 4.5.

4.4.1 Sampling for quality conformance inspection.

4.4.1.1 Complete machines (with scales).

4.4.1.1.1 Inspection lot. Drafting machines of the same type and style, complete with case if applicable, offered for delivery at one time, shall be considered a lot for purposes of acceptance inspection.

4.4.1.1.2 Sampling for examination. A random sample of machines shall be selected from each inspection lot by the inspector in accordance with inspection level II of MIL-STD-105. The acceptable quality level (AQL) shall be 1.5 percent defective.

4.4.1.1.3 Sampling for tests. A random sample of machines shall be selected from each inspection lot in accordance with inspection level S-3 of MIL-STD-105. The AQL shall be 1.5 percent defective. Machines, and if applicable, the cases in the sample, shall be subjected to the tests specified in 4.5.1.

4.4.1.1.4 Sampling for inspection of packaging. Sample packages and packs shall be selected and inspected in accordance with MIL-P-116 to conform to the requirements of section 5 herein.

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4.4.1.2 Scales only.

4.4.1.2.1 Inspection lot. Scales offered for delivery at one time shall be considered a lot for purposes of inspection.

4.4.1.2.2 Sampling for examination. A random sample of scales shall be selected from each lot in accordance with MIL-STD-105 at inspection level II. The AQL shall be 1.5 percent.

4.4.1.2.3 Sampling for tests. A random sample of scales shall be selected from each lot in accordance with MIL-STD-105 at inspection level II. The AQL shall be 1.5 percent defective. Scales in the sample shall all be subjected to the tests specified in 4.5.2.

4.4.1.2.4 Sampling for inspection of packaging. Sample packages and packs shall be selected and inspected in accordance with MIL-P-116 to conform to the requirements of section 5 herein.

4.5 Examination and test procedures. Each of the sampled equipment shall be examined or tested as specified herein to verify compliance with this specification. Any equipment in the sample containing defects shall be rejected, and if the number of defective equipments in the sample exceeds the acceptance number for that sample, the lot represented by the sample shall be rejected.

4.5.1 Drafting machine and scales.

4.5.1.1 Examination. Each sample selected in accordance with 4.4.1.1.2 shall be given a visual and dimensional examination.

4.5.1.2 Tests. Samples for tests shall be selected in accordance with 4.4.1.1.3.

4.5.1.2.1 Parallelism test. Attach the drafting machine to a drafting board of sufficient size to construct the lines shown on figure 1 on a sheet of drawing paper. The machine is to be attached and operated in the manner as is intended for normal use. The machine shall have an 18-inch scale, which meets all the provisions of this specification, attached to the horizontal scale arm of the machine. Align and lock the scale arm in a horizontal position with one of the witness points alined with the 0 degree protractor marking. Release the protractor lock and rotate the scale arm 90 degrees counterclockwise such that the scale is in a vertical position. Using a sharp pencil, mark the paper at the 0- and 18-inch scale graduation points. Identify the mark closest to the protractor head as "A". Release the protractor lock and rotate the scale arm clockwise 90 degrees, such that the scale is returned to the original horizontal position. Draw a horizontal line from point "A" for a length of 18 inches and mark the end of the line as "B". Draw a horizontal line 18 inches above line "AB" through the previously marked point. Release the protractor lock and rotate the scale arm counterclockwise 90 degrees, such that the scale is in the vertical position. Draw vertical lines upward from points "A" and "B" through the upper horizontal line and label the intersection points "C" and "D", respectively. Measure lines "BD" and "CD" to the nearest 0.01 inch. If the length of either line "BD" or "CD" is longer than 18.08 inches or shorter than 17.92 inches, the drafting machine shall be considered defective.

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4.5.1.2.2 Indexing accuracy. Drafting machines equipped with either a 3- or 15-degree automatic indexing feature shall be secured to a drafting table and tested as follows: Mount both scales in the scale arm and adjust the scales so that the angle between lines drawn along the edge of the scales is 90 degrees. By means of the indexing mechanism, draw a line every 3- or 15-degrees within the 90 degree angle; check for accuracy of each angle so constructed. The angles shall be accurate to within plus or minus 10 minutes.

4.5.1.2.3 Protractor accuracy. The protractor shall be secured to a drafting table and tested as follows: The protractor shall be set for a predetermined baseline setting and a line drawn at this setting. Rotate the scale clockwise or counterclockwise through 45 degrees either side of the baseline, and check the respective angles for accuracy. The accuracy of the angles shall be within plus or minus 10 minutes.

4.5.2 Scales only.

4.5.2.1 Examination. Each sample selected in accordance with 4.4.1.2.2 shall be given a visual and dimensional examination. To determine conformance to 3.4.3.1, the spacing of the graduation lines shall be checked by means of a pair of dividers and a graduated test bar of known accuracy.

4.5.2.2 Tests. Samples for tests shall be selected in accordance with 4.4.1.2.3.

4.5.2.2.1 Scale flatness test. The degree of flatness of the scales shall be tested by placing the underside of the scale upon a firm, level, nonresilient surface and determining the degree of longitudinal bow by means of a feeler gage. No weights shall be applied to the scales during this test. If the degree of longitudinal bow exceeds the limits specified in 3.6.1, the scale shall be rejected.

4.5.2.2.2 Straightness of scale test. The straightness of scales shall be tested as follows: Securely fasten a good grade of drawing paper on the drawing board. Drive two identical pins into the drawing board, about 1/4-inch deep and spaced approximately 1/2-inch less than the length of the scale to be tested. The pins shall be driven vertically into the drawing board. Place the edge of the scale being tested against the pins and draw a line along the edge with a well-sharpened, hard-lead pencil. Rotate the scale 180 degrees, place the same edge against the other side of the pins and draw another line along the edge. Measure the distance between the two lines, at increments not exceeding 3 inches, using a shop microscope or other precision method. If the variation in distance between the two lines exceeds the value specified in 3.6.2, the scale shall be rejected. Repeat the above test for the opposite drawing edge of the scale.

4.5.2.2.3 Parallelism test. The parallelism of the edges of the scale shall be tested as follows: Place the scale over a piece of drawing paper with the chuck plate on the upper surface. Draw a line along each outside edge of the scale using a well-sharpened, hard-lead pencil. Remove the scale from the paper and measure the distance between the two lines, at increments not exceeding 3 inches, using a shop microscope or other precision method. If the variation in distance between the two lines exceeds the value specified in 3.6.2 the scale shall be rejected.

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4.6 Tests.

4.6.1 High impact. The first article model type I drafting machine (less scales) shall be tested in accordance with the following provisions of MIL-S-901:

Shockproof - Grade B
Equipment classification - Subassembly.
Classes - Class I.
Test classification - Lightweight.
Type - Type C.
Mounting - 250 pound weight shall be used with test fixture shown in figure entitled "Shock testing machine for light weight equipment".
Mode - Machine shall be attached to the test platform in the same manner as attached on a plotting table with the head in the stored position.
Disposition - Delivered in the same manner as all items on contract.
Quantity - One drafting machine.

4.6.2 Storage. The drafting machine (less scales) furnished as the first article model shall be placed in a temperature-controlled cabinet and subjected to an ambient temperature of 71 ± 1.4°C, with at least 50 percent relative humidity for a period of 4 hours. The temperature shall then be gradually lowered to minus 62 ± 1.4°C for a period of 72 hours. At the end of this cycle, the drafting machine shall be allowed to return to room temperature and shall be examined for defects. Presence of any defects which affect serviceability or appearance of the drafting machine shall be cause for rejection.

4.7 Rejected lots. If an inspection lot is rejected, the contractor may withdraw the lot, rework it to correct the defects, or screen out the defective units, as applicable and reinspect. Such lots shall be separate from new lots and shall be clearly identified as reinspected lots.

4.8 Packaging inspection. Sample packages and packs and the inspection of the packaging, packing and marking for shipment and storage shall be in accordance with the requirements of section 5 and the documents specified therein.

5. PACKAGING

(The preparation for delivery requirements specified herein apply only for direct Government acquisition. For the extent of applicability of the preparation for delivery requirements of referenced documents listed in section 2, see 6.5.)

5.1 Preservation and packaging. Packaging shall be level A or C as specified (see 6.2.1).

5.1.1 Level A - complete drafting machine. Each drafting machine and accessories shall be cleaned prior to packaging. When a carrying case is furnished, the drafting machine and accessories shall be secured in place within the case. One complete machine, together with accessories, shall be preserved

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and packaged in accordance with MIL-P-116, method IA and placed in a snug fitting fiberboard box conforming to PPP-B-636, type CF, class-weather resistant, variety SW, grade W5c or W6c, style RSC. Box closure shall be method V as specified in the appendix of PPP-B-636. Cushioning of the wrapped carrying case within the unit container shall be accomplished by applying a minimum 1-inch thick sheets of unicellular polyethylene cushioning conforming to PPP-C-1752, type II, class 1, on all surfaces. Scales which are furnished as an accessory and are not secured within the carrying case shall be packaged in accordance with 5.1.2.

5.1.2 Level A - scales only. Each scale shall be unit protected by a non-abrasive material in accordance with MIL-P-116, method III. Contact preservatives will not be required. Scales shall be individually packaged in containers conforming to any of the following specifications at the option of the contractor.

Specification	Box
PPP-B-566	Folding, paperboard
PPP-B-676	Set-up, paperboard

Container closure shall be in accordance with the applicable container specification or appendix thereto.

5.1.3 Level C. Drafting machines or scales shall be preserved and packaged in accordance with the contractor's commercial practice to assure carrier acceptance and adequate protection against physical damage during shipment from the supply source to the first receiving activity.

5.2 Packing. Packing shall be level A or C as specified (see 6.2.1).

5.2.1 Level A. Drafting machines or scales, preserved and packaged in accordance with 5.1 and as specified (see 6.2.1), shall be packed in fiberboard boxes conforming to PPP-B-636, type CF, class weather-resistant, variety SW, grade W5c or W6c, style RSC. Box closure and reinforcement shall be accomplished in accordance with the appendix of PPP-B-636.

5.2.2 Level C. Drafting machines or scales, preserved and packaged in accordance with 5.1 and as specified (see 6.2.1), shall be packed in accordance with the contractor's commercial practice to assure carrier acceptance and acequate protection against physical damage during shipment from the supply source to the first receiving activity. Minimum requirements shall be in compliance with Uniform Freight Classification Ratings, Rules and Regulations.

5.3 Marking. In addition to any special marking specified in the contract or herein, each unit and intermediate package and shipping container shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. Drafting machines furnished under this specification are intended for navigational plotting and drafting (special and general) applications.

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6.2 Ordering data.

6.2.1 Acquisition requirements. Acquisition documents should specify the following:

(a) Title, number and date of this specification.
(b) Type, class and style required (see 1.2).
(c) Quantity of items required - machines with scales or scales only.
(d) Whether sample for first article inspection is required (see 3.2).
(e) In what style general drafting machine shall be furnished (see 3.4.2).
(f) Whether micrometer adjusting screw is required for type II, class 2 drafting machines (see 3.4.2.4.2).
(g) Degree interval of automatic indexing (see 3.4.2.5).
(h) Whether mounting shall be for wooden or metal table or board (see 3.4.2.6).
(i) Arm size (see 3.4.2.7).
(j) Whether counterbalance is required (see 3.4.2.8).
(k) When letters and numerals shall be 5/64-inch high (see 3.4.3.1).
(l) Whether carrying case is required (see 3.8.2).
(m) Level of packaging and packing required (see 5.1 and 5.2).
(n) Class of fiberboard box required (see 5.2.1).

6.2.2 Data requirements. When this specification is used in an acquisition which incorporates a DD Form 1423, Contract Data Requirements List (CDRL), the data requirements identified below shall be developed as specified by an approved Data Item Description (DD Form 1664) and delivered in accordance with the approved CDRL incorporated into the contract. When the provisions of FAR 52.227-7031 are invoked and the DD Form 1423 is not used, the data specified below shall be delivered by the contractor in accordance with the contract or purchase order requirements. Deliverable data required by this specification is cited in the following paragraph.

Paragraph no.	Data requirement title	Applicable DID no.	Option
4.3.1	First article inspection report	DI-T-4902	----

(Data item descriptions related to this specification, and identified in section 6 will be approved and listed as such in DoD 5000.19L., Vol. II, AMSDL. Copies of data items descriptions required by the contractors in connection with specific acquisition functions should be obtained from the Naval Publications and Form Center or as directed by the contracting officer.)

6.2.2.1 The data required of 6.2.2 and any task in section 3, 4, or 5 of this specification required to be performed to meet a data requirement may be waived by the contracting/acquisition activity upon certification by the offeror that identical data were submitted by the offeror and accepted by the Government under a previous contract for identical item acquired to this specification. This does not apply to specific data which may be required for each contract regardless of whether an identical item has been supplied previously (for example, test reports).

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6.3 First article inspection. Invitations for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection as to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract.

6.4 Machine arm size. As a guide to selection of arm size, the following is the intended size arms to be used with various size drafting boards.

16-inch arms - board 24 inches by 36 inches
18-inch arms - board 27 inches by 40 inches
24-inch arms - board 36 inches by 60 inches
30-inch arms - board 42 inches by 84 inches
36-inch arms - board 48 inches by 96 inches

6.5 Sub-contracted material and parts. The preparation for delivery requirements of referenced documents listed in section 2 do not apply when material and parts are acquired by the contractor for incorporation into the equipment and lose their separate identity when the equipment is shipped.

6.6 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

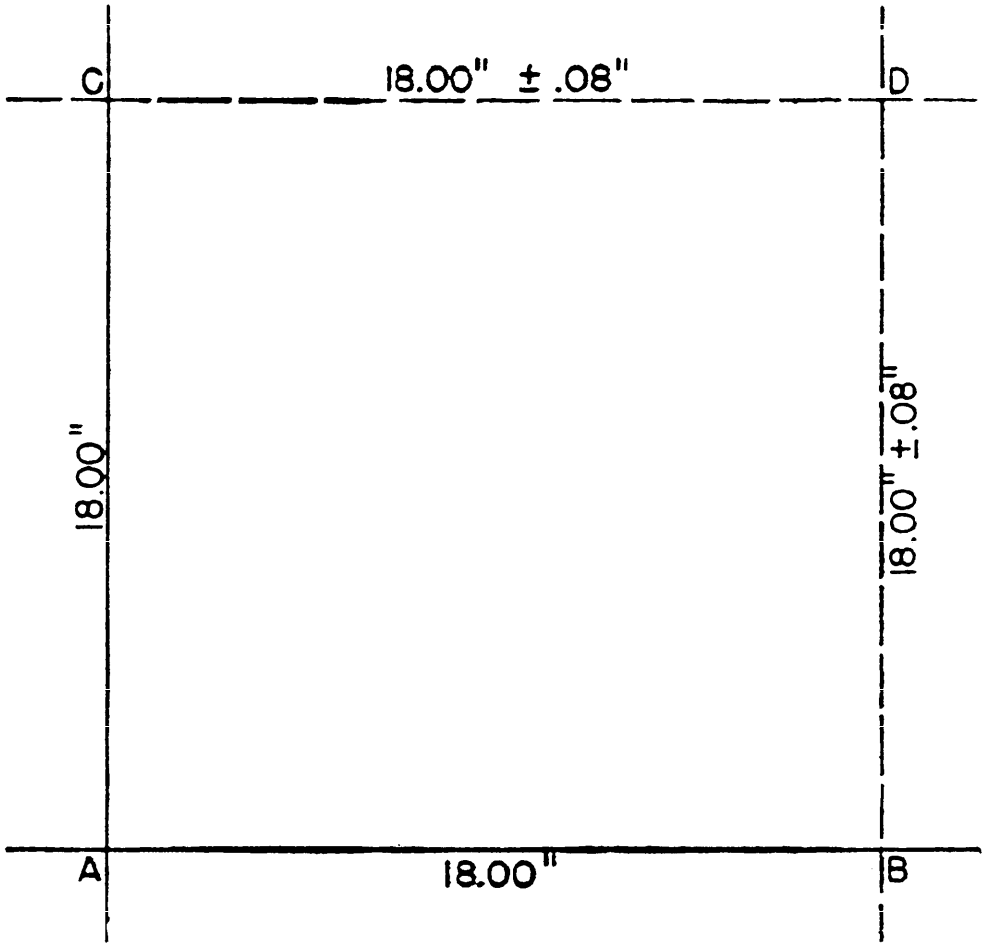
Custodians:
Navy - SH
Air Force - 99

Review activities:
Navy - MC
Air Force - 82
DLA - GS

User activity:
Army - ME, AT

Preparing activity:
Navy - SH
(Project 6675-0745)

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SH 131817

FIGURE 1. Parallelism test diagram.

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-D-3898B	DRAFTING MACHINE AND SCALES GENERAL SPECIFICATION FOR

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*:

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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